FRAME 1

# DECISION

*What decision are we trying to make?*

## Planning question

How much Honda City 1.2L demand should be planned for over the next 12 weeks — and how much should be committed now versus kept flexible?



## Recommended planning posture

**848**
base expected demand

**750**
firm commitment

**185**
flexible option

**935** **protected planning capacity**

covers the +10% upside case (~933 units)

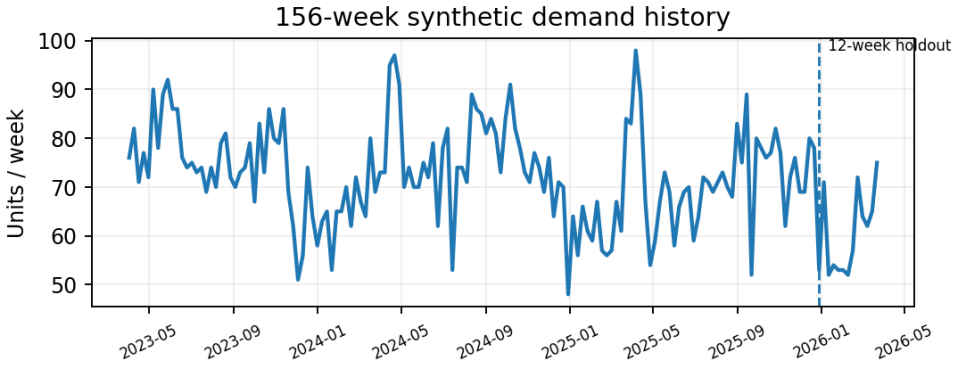
**Business consequence**

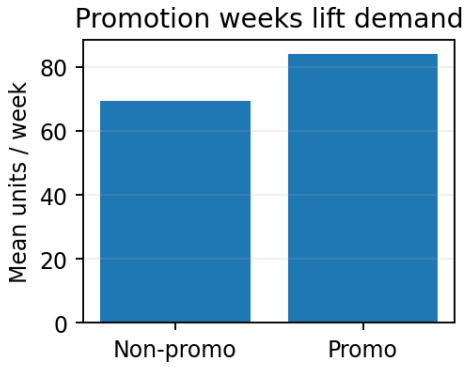
Secure the base volume without locking the full upside risk into firm capacity.

FRAME 2

# DEMAND PATTERN

*What is happening in demand?*




| 71.7 | 14.5% | 84.2 | 69.4 |
|---|---|---|---|
| mean units/week | CV | promo-week mean | non-promo mean |

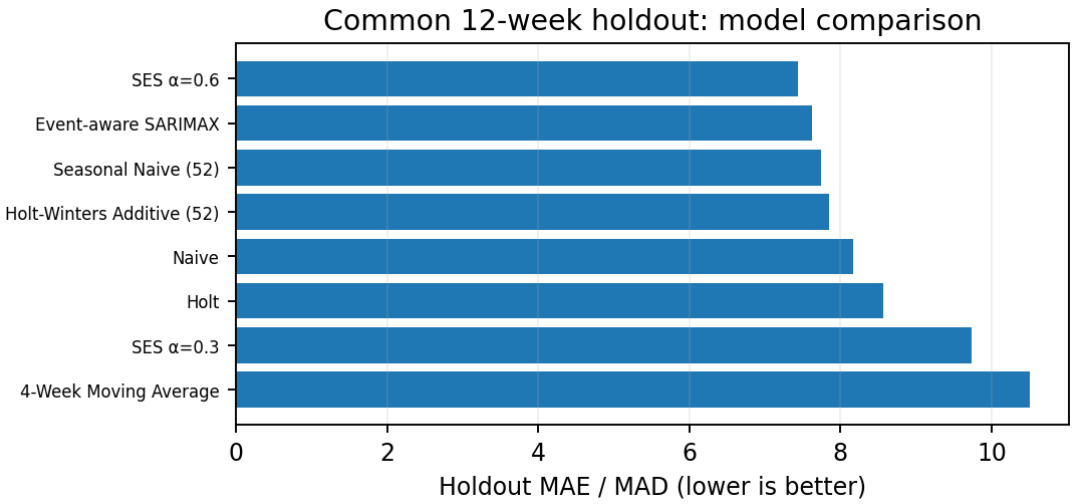
**Key insight**

The series is level-dominated with mild downward drift, short event/holiday peaks, variable promotion uplift and isolated stockout suppression. A responsive smoothing method is therefore more defensible than a heavy trend model.

FRAME 3

# MODEL CHOICE

*Which method earned our trust?*




## SES α = 0.6

**Selected on the same unseen 12 weeks**

Lowest MAE, RMSE and WAPE; MAPE remains close to the best baseline.

| 7.4 | 12.6% | –3.0 |
|---|---|---|
| MAE | MAPE | Tracking Signal |

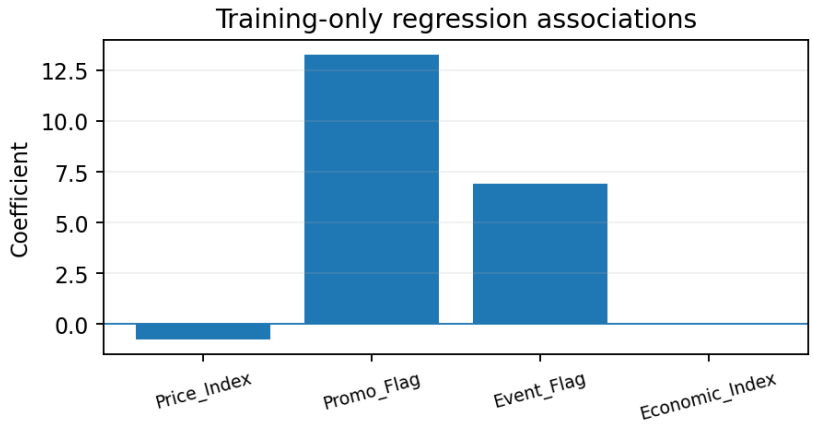
**Why not Naive?**

Naive had strong MAPE, but TS = +11.5 indicates persistent under-forecasting.

FRAME 4

# DRIVERS

*What appears to move demand?*




## Training-only regression

Demand ~ Price Index + Promotion Flag + Event Flag + Economic Index

**0.463**

**0.448**

**0.885**

**Signal** Price association is negative; promotion and event flags are positive. These are associations, not proof of causation.

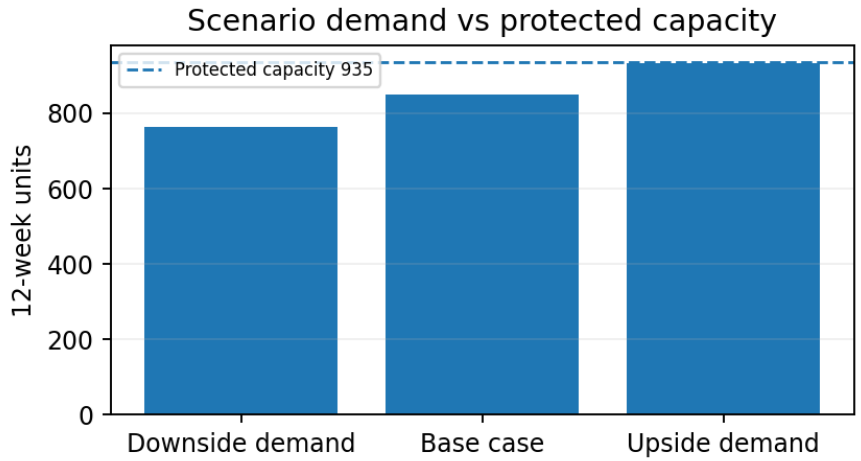
### Omitted-variable caveat

Dealer bookings, financing approvals, competitor pricing, dealer inventory and vehicle availability may also shape observed demand.

FRAME 5

# RISK

*Where could the forecast fail?*




**BIAS / TS**

**–3.0**

negative = over-forecasting

**AVAILABILITY SHOCK**

**–15%**

≈721 serviceable units

**Stress range**

Downside 763 | Base 848 | Upside 933 units over 12 weeks.

Largest misses include event effects, stockout suppression and short-term variation.

**Management interpretation**

The model is useful, but it cannot fully predict sudden availability or event shocks. Flexibility is the risk-control mechanism.

FRAME 6

# ACTION

*What should management do now?*

## Recommended 12-week posture

# 750 + 185

**firm commitment + flexible option**

Protected planning capacity = 935 units. This covers the +10% upside scenario of approximately 933 units without committing the entire upside volume upfront.

**12-WEEK CAPACITY LOGIC**




### Review trigger

Reforecast if actual demand differs from plan by >10% for two consecutive weeks OR Tracking Signal shows sustained drift.

### Key risk

Stockouts and abrupt event-driven demand changes can move actual demand outside the smooth forecast path.

**Decision in one sentence:** Plan around 848 expected units, lock 750, preserve 185 of flexible capacity, and reforecast quickly when the trigger is breached.